BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions) (43)

Maximum Marks: 43

Note: This section consists of 43 part questions and all are to be answered. Each question carries ONE mark. Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 2 Hours — Maximum Marks: 85 (As per condensed syllabus)

Q1. Choose the correct answer for each from the given options:

Qi. Which pair of elements belongs to same group in the periodic table:

(A) Mg and S
(B) O and Br
(C) Na and Al
(D) C and Si

Qii. Hydride ion and Helium atom have the same:

(A) Number of protons
(B) Number of neutrons
(C) Valency
(D) Number of electrons

Qiii. The valence shell electronic configuration of Halogen family is:

(A) ns^1
(B) ns^2, np^6
(C) ns^2, (n-1)d5
(D) ns^2, np^5

Qiv. Dolomite is a carbonate of:

(A) Ca and Ba
(B) Li and Na
(C) Mg and Al
(D) Mg and Ca

Qv. Nitric acid is frequently transported in a container made of :

(A) Al
(B) Cu
(C) Zn
(D) Ag

Qvi. The mixture of Aluminium powder and Aluminium nitrate is called:

(A) Gun powder
(B) Ammonal
(C) Urea
(D) Tincal

Qvii. 'A' is the element of IIA group while 'B' belongs to VA group. When 'A' reacts with 'B' they :

(A) AB
(B) AB_3
(C) A_1B_2
(D) A_2B_3

Qviii. Diaspore is an ore of:

(A) Copper
(B) Zinc
(C) Iron
(D) Aluminium

Qix. Metallic character of p-block elements depends upon electron population of outer most shell and:

(A) Ionization potential
(B) Shielding effect
(C) Electronegativity
(D) Hydration energy

Qx. Which of the following ligand is chelating agent:

(A) EDTA
(B) CN^{-1}
(C) NH_3
(D) F^{-1}

Qxi. The formula of chromite ore is:

(A) CrO_2Cl_2
(B) $K_2Cr_2O_3$
(C) $FeOCr_2O_3$
(D) K_2CrO_3

Qxii. The Coordination number of Cobalt in $[Co(en)_2Cl_2)Cl$ is:

(A) 3
(B) 4
(C) 6
(D) 8

Qxiii. The element which is present in group IV-A and 3rd period, its atomic number is:

(A) 15
(B) 14
(C) 13
(D) 6

Qxiv. Oxidation state of oxygen in super oxides is:

(A) -2
(B) +2
(C) -1/2
(D) -1

Qxv. 'Li' is powerful reducing agent due to its large value for:

(A) Ionization potential
(B) Hydration enthalpy
(C) Electron Affinity
(D) Electronegativity

Qxvi. Aluminium does not react with:

(A) H_2SO_4 (B) H_3PO_4

(C) HNO_3 (D) HCl

Qxvii. The ligands which form ring structure with central atom is:

(A) Chelate (B) Co-ordination sphere

(C) Cyclic ligand (D) Polydentate ligand

Qxviii. Aluminium bronze contains:

(A) 10% Al and 90% Cu (B) 5% Al and 95% Cu

(C) 20% Al and 80% Cu (D) 90% Al and 10% Cu

Qxix. The number of unpaired electrons in ferrous ion is:

(A) 2 (B) 3

(C) 4 (D) 6

Qxx. Coating of Zinc on iron is called:

(A) Surface coating (B) Electroplating

(C) Galvanizing (D) Sheardizing

Qxxi. On jumping from eg to t_2g orbitals electron radiate energy in this region of spectrum:

(A) ultraviolet (B) visible

(C) infrared (D) far infrared

Qxxii. Alkanes are also called:

(A) olefins (B) paraffins

(C) proteins (D) acetylene

Qxxiii. Calcium Carbide on hydrolysis forms:

(A) methane (B) ethane

(C) benzene (D) acetylene

Qxxiv. This gas is used in welding:

(A) Methane (B) Ethane

(C) Ethene (D) Ethyne

Qxxv. Which of the following gives electrophilic substitution reaction:

(A) Benzene (B) Formaldehyde

(C) Ethanol (D) Ethyne

Qxxvi. This is the most stable carbonium ion:

(A) CH_3 (B) RCH_2

(C) R_2CH (D) R_3C

Qxxvii. A dipolar charged but overall electrically neutral ion is called:

(A) Carbinium ion (B) Zwitterion

(C) Oxonium ion (D) Carbanion ion

Qxxviii. This imparts green colour to glass:

(A) Cr_2O_3 (B) CuO

(C) CoO (D) MnO_2

Qxxix. Glucose and Fructose:

(A) Position Isomers (B) Chain Isomers

(C) Functional Group Isomers (D) Metamers

Qxxx. Which one of the following is Acidic Amino Acid:

(A) Glycine (B) Alanine

(C) Arginine (D) Aspartic acid

Qxxxi. Wood spirit is the common name of:

(A) Ethanol (B) Methanol

(C) Starch (D) Sugar

Qxxxii. Which one of the following is a heterocyclic compound:

(A) Pyridine (B) Phenol

(C) Benzene (D) Methane

Qxxxiii. The animal starch is:

(A) Cellulose (B) Glycogen

(C) Amylose (D) Glucose

Qxxxiv. The compound obtained by the dry distillation of calcium formate is:

(A) Methanol (B) Methanal

(C) Methane (D) Acetone

Qxxxv. Another name of fructose is:

(A) d(+) Glucose (B) Levulose

(C) Oligosaccharide (D) Galactose

Qxxxvi. Bakelite is a polymer of phenol and :

(A) Formaldehyde
(B) Ethanol
(C) Ethene
(D) Benzene

Qxxxvii. Dehydration of Ethanol produces:

(A) Ethyne
(B) Ethane
(C) Ethene
(D) Water gas

Qxxxviii. The hybridisation in carbonyl group is:

(A) sp
(B) sp^2
(C) sp^3
(D) dsp^2

Qxxxix. Polyhydric aldehydes or ketones are known as:

(A) Carbohydrates
(B) Protein
(C) Enzymes
(D) Glass

Qxl. The number of acidic hydrogen present in 1-butyne:

(A) 1
(B) 2
(C) 3
(D) 4

Qxli. Which one of the following meta directing group:

(A) $-NH_2$
(B) $-CH_3$
(C) $-NO_2$
(D) -OH

Qxlii. Glyoxal is formed by ozonolysis of:

(A) ethene
(B) benzene
(C) acetone
(D) acetylene

Qxliii. When CO_2 reacts with Grignard reagent it give:

(A) Alcohol
(B) Carboxylic acid
(C) Alkane
(D) Ester

Section 'B' (Short-Answer Questions) (42)

Maximum Marks: 42

Note: Attempt FIVE questions from this section in all, selecting at least two questions from Inorganic chemistry and two from organic chemistry. All questions carry equal marks.

QINORGANIC CHEMISTRY.

Q2.

(i) Refer to the list of given compounds. Compound: A = Carnallite, B = Blue Vitriol, C = Potash Alum, D = Barytes. * Write chemical formula of A and D. * Write chemical formula of B and give chemical equation when it is heated at 230°C. * Write three users of C.

(ii) Write the IUPAC names of any four of the following: * $K_3[Fe(C_2O_4)_3]$ * $[Co(CN)_6]^{-3}$ * $Cu(NH_3)_4SO_4$ * $[Ni(en)_2Cl_2]^{+2}$ * $[Cr(H_2O)_6](NO_3)_3$

(iii) Write the group and periods of the following elements with Atomic numbers. * Z = 18 * Z = 24 * Z = 34 * Z = 47

(iv) Write equation for the following reactions: * Reaction of SO_2 with Cl_2 * Heating MgO with carbon and Cl_2 gas * Reaction of Mg with cold and dilute HNO_3 * Reaction of Aluminium with conc H_2SO_4

(v) Describe the extraction of sodium metal by Down's process.

(vi) What is Aqua regia? How does gold dissolved in it. Write its equation.

QORGANIC CHEMISTRY.

(vii) Define the following: * Polymerization * Homologous series * Glycosidic Linkage * Catenation

(viii) Write the classification of organic compound with examples.

(ix) The structure of two significance organic molecules A and B are shown below: A = ethane (H_3C-CH_3, all H atoms drawn), B = ethene ($H_2C=CH_2$, all H atoms drawn). * Draw the hybrid orbital structure of 'B' * Give simple chemical test between A and B * Write equation for the preparation of 'A' from CH_3I. * Write chemical equation for reaction of 'B' with S_2Cl_2.

(x) Write I.U.P.A.C names any four of the following: * $(CH_3)_2 - CH - (CH_2)_2 - CH(CH_3)_2$ * $CH_3 - (CHOH)_2 - CH_2OH$ * $CH_2 = CHCHClCH_2C \equiv CH$ * $CH_2 = CH - CH_2CHO$ * $CH_3 - CH_2COCH(CH_3)_2$

Section 'C' (Detailed-Answer Questions) (17)

Maximum Marks: 17

> Note: Answer 2 questions from this Section. One from Inorganic Chemistry and one from Organic Chemistry.

QINORGANIC CHEMISTRY.

Q3.

What is metallurgy? Describe the extraction of 99.99% pure Aluminium from bauxite one containing SiO_2 and Fe_2O_3 as impurities.

OR Complete and balance the following equations: * $Al_4C_3 + H_2O \rightarrow$ * $K + O_2 \rightarrow K + O_2$ * $Cr_2O_3 + KOH + Br_2 \rightarrow$ * $CuSO_4 + KI \rightarrow$ * $Al + KOH + H_2O \rightarrow$ * $S + HNO_3 \rightarrow$ * $CH_4 + HNO_3 \rightarrow$ * $Fe_2O_3 + Al \rightarrow$

Q4.

Define Binary compounds of hydrogen. Give their classification. Explain the preparation and properties of ionic hydride.

QORGANIC CHEMISTRY.

Q5.

Explain the molecular orbital treatment of benzene; write the reaction of benzene with H_2 and O_2.

OR Explain SN^1 and SN^2 reaction mechanism with example.

Q6.

Define Fermentation. How is ethanol manufacture by the fermentation of the following. * Starch * Molasses

--OR Complete the following reactions: * $C_2H_5OH + SOCl_2 \text{—(propane)}\rightarrow$ * $CH_3 - CH_2OH \text{—(Cu-Ni / 180°C)}\rightarrow$ * $CH_3 - C(=O) - CH_3 + H_3 \text{—(Pd)}\rightarrow$ * $CH_3 - C(=O) - CH_3 + 3I_2 + 2Na_2CO_3 \rightarrow$ * $HCHO + Cu(OH)_2 + NaOH \rightarrow$ * $HC \equiv CH + [O] \text{—(KMnO}_4\text{ / Cold)}\rightarrow$ * $C_2H_5OH + CH_3COOH \text{—(}H_2SO_4\text{)}\rightarrow$ * $CH_3 - CH(Cl) - CH_3 + Na \rightarrow$ (2-chloropropane structure: Cl below the middle carbon) * $CH_3COOH \text{—(}MnO_2\text{ / 500°C)}\rightarrow$

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